Chapter 35 Using the keyboard

This chapter shows how to use the keyboard, for text and control input, and how to control its behaviour.

35.1 Standard facilities

35.1.1 Keyboard layout

The standard **Hunter 16** keyboard layout is largely compatible with the UK version of the 83 key IBM PC keyboard, but with minor changes to the layout and ways that certain less frequently used key sequences are obtained. International keyboard layouts are catered for by the use of the INTKEYB utility.

35.1.2 BIOS

The standard BIOS services, provided by INT 16H, are as follows:

00H	Get key
01H	Get keyboard buffer status
02H	Get shift status

35.1.3 Interrupts

The keyboard generates the following interrupts:

INT 9	- when a key is pressed or released
INT 1BH	- when the break key is pressed (passed via BIOS)
INT 23H	- when the break key is pressed (passed via MS-DOS)

When a key is pressed, the keyboard generates an INT 9 interrupt together with a 'scan code' (in the range 1..83 decimal) which can be read by the INT 9 service routine to identify the key. When a key is released, another INT 9 is generated together with the same scan code, with bit 7 set (ie 128 + 1..83 decimal).

The BIOS INT 9 service routine responds to the first INT 9 by translating the scan code into two bytes which it places in the keyboard buffer. If f the key is a character key, the first byte is the corresponding

ASCII code (01..FFH) and the second is the scan code. If the key is a 'special key (for example, [F1]), the first byte is 00H and the second (auxiliary) byte identifies the key.

A program can also replace the processing of scan codes (and hence all keyboard input) by changing the INT 9H vector to point to its own code.

For details of scan codes, see Appendix E.

The break interrupts are not used by the BIOS, so a program can respond to the break key by changing the corresponding vector (see section 18.1) to point to its own code.

35.1.4 RAM

RAM location 00417H provides information about the keyboard state:

Bit	Significance if set
7	Insert mode
6	Caps lock mode
5	Num lock mode
4	Scroll lock mode
3	[Alt] active
2	[Ctrl] active
1	Left [Shift] active
0	Right [Shift] active

RAM location 00418H provides information about the shift keys:

Bit	Significance if set
7	[Ins] pressed
6	[Caps Lock] pressed
5	[Num Lock] pressed
4	[Scroll Lock] pressed
3	[Ctrl][Num Lock] pressed
2	not used
1	not used
0	not used

35.2 Extra facilities

35.2.1 Extra keys

The Hunter 16 keyboard includes two 'extra' keys [PWR] (used to turn on and off) and [PAW] (used to control Hunter 16 hardware, such as the LCD, backlight and sound). (For further details of the controls, see Part 2.)

These extra keys are decoded in a quite different manner to the standard IBM keys: pressing them does not generate scan codes, so they cannot be detected directly from an INT 9 service routine, nor emulated by inserting codes in the keyboard buffer.

To simulate pressing one of these keys, you can access the appropriate control routine directly. For example, to simulate pressing (PWR), you could use INT 15H, function 68H; to simulate pressing the backlight key, use INT 10H function 64H.

To redefine these keys (or assign their functions to other keys), use INT 16H, function 24H.

35.2.2 BIOS

The Hunter 16 BIOS provides the following additional services:

Typing keys	Set key repeat	INT 16H, function 03H
	Set key repeat rate/delay	INT 16H, function 29H
	Control key repeat	INT 16H, function 2AH
	Control keyclick	INT 16H, function 26H
	Write to buffer	INT 16H, function 5H
Read keys	Extended get key	INT 16H, function 10H
	Extended get status	INT 16H, function 11H
	Extended get shift status	INT 16H, function 12H
Shift keys	Set temporary shift	INT 16H, function 20H
	Control shift keys	INT 16H, function 24H
[Prtsc] key	Control screen dump area	INT 16H, function 27H
Reboot and break keys	Control [Ctrl][Alt][DEL]	INT 16H, function 22H
	Control emergency breakout	INT 16H, function 23H
	Control BREAK keys	INT 16H, function 2DH

Redefine keys	Redefine key codes	INT 16H, function 24H
	Redefine key scan codes	INT 16H, function 2BH
	Redefine [PAW] keys	INT 16H, function 2CH
	Reset keyboard	INT 16H, function 25H
	Define power-up keys	INT 15H, function 71H
	Read power-up keys	INT 15H, function 88H
	Disable HOT key	INT 15H, function 7BH
	Disable HOT key power facilities	INT 15H, function 7CH
Simulate extra keys	Power down	INT 15H, function 68H
	Move LCD window	INT 10H, function 74H
		INT 10H, function 76H
	Set contrast	INT 10H, function 63H
	Set backlight	INT 10H, function 64H
	HOT key	INT 15H, function 7AH

The event-handling mechanism also provides a convenient way to detect whether certain keys have been pressed ([Ctrl-c], [Break], [PWR]) etc). See Chapter 41 for more details.

35.3 Using the extra facilities

35.3.1 Redefining keys

You can redefine any (or all) of the IBM-standard keys by using INT 16H function 2BH. (To redefine the extra Hunter 16 keys, use INT 16H function 24H.)

If another application might have been run on the unit which also redefines keys (in different ways), you should reset the keyboard layout first (INT 16H function 25H), so you know where you are!

To redefine a key, you need to know its normal key code and the key code for the key that you want it to emulate. This information is provided in Appendix E. To disable a key entirely, redefine it as key 00H.

For example, if an application requires a lot of numeric input, you could redefine the number keys in the numeric keypad as the main number keys, to save the user having to use Num Lock :

```
set A to 25H   ;RESET KEYBOARD FUNCTION
INT 16H        ;EXECUTE FUNCTION

set AH to 2BH  ;DEFINE KEY FUNCTION
set AL to 52H  ;KEY PAD 0 KEY
set BH to 0BH  ;MAIN 0 KEY
INT 16H        ;EXECUTE FUNCTION

set AL to 4FH  ;KEY PAD 1 KEY
set B to 2     ;MAIN 1 KEY
INT 16H        ;EXECUTE FUNCTION

set AL to 50H  ;KEY PAD 2 KEY
set BH to 3    ; MAIN 1 KEY
INT 16H        ;EXECUTE FUNCTION

etc, etc
```

(You could achieve the same result rather more easily by forcing NUM LOCK mode by using INT 16H function 21H.)

Remember that if you redefine a key, it will type a different character to the one on its key cap, which might confuse users somewhat! If you use this facility, warn users first.

35.3.2 Protecting an application

When you install an application on **Hunter 16**, you may prefer for that unit to be used only to run that application. This might be necessary, for example, if the unit will be used by personnel who tend to 'break' applications.

To do this, you must disable the [Ctrl][Alt][DEL] and emergency breakout facilities, by including the following code near the start of the application:

```
set AH to 22H  ;CONTROL REBOOT FUNCTION
set AL to 1    ;DISABLE OPTION
set BH to 7    ;SET PARAMETERS TO ALLOW FUNCTION
set BL to 8
set CH to 9
set CL to 10H
set DH to 11H
set DL to 12H
INT 16H        ;EXECUTE FUNCTION
```

```
set AH to 23H  ;CONTROL BREAKOUT FUNCTION
INT 16H        ;EXECUTE FUNCTION
```

Note that after this, to leave the application you will either need the help of a special dongle (held by your supplier), or have to remove the main batteries for several weeks (which loses all data stored in memory and on the RAM disk).

To minimise problems, you should include a password-protected 'exit to DOS' option in the application.

35.3.3 Initialising the keyboard

At the start of an application, you may want to be sure of the keyboard state. For example, if Hunter 16 had been running another application, it might have redefined some keys or disabled certain shift keys. Even if another application has not been running, the user might have engaged keys such as Num Lock before running the program, which will affect behaviour of the keyboard.

Include the following code at the start of an application to ensure that the keyboard is always initialised to the same state (the same as a cold start):

```
;*************************************
;* EXAMPLE: INITIALISING THE KEYBOARD *
;*************************************

;FORCE NORMAL KEYBOARD (ALSO OVERRIDES ANY REDEFINED KEYS)

set AH to 28H  ;KEYBOARD TYPE FUNCTION
set AL to 1    ;INSTALL NORMAL KEYBOARD
INT 16H        ;EXECUTE FUNCTION


;ENABLE ALL SHIFT KEYS, AND SET SHIFTS TO INACTIVE

set AH to 21H  ;CONTROL SHIFT KEYS FUNCTION
set AL to 0    ;ENABLE ALL KEYS
set BL to 0    ;RESET ALL SHIFTS
INT 16H        ;EXECUTE FUNCTION
```

```
;ENABLE REBOOT

set AH to 22   ;CONTROL REBOOT FUNCTION
set AL to 0    ;ENABLE OPTION
set BH to 7    ;SET PARAMETERS TO ALLOW FUNCTION
set BL to 8
set CH to 9
set CL to 10H
set DH to 11H
set DL to 12H
INT 16H        ;EXECUTE FUNCTION


;ENABLE EMERGENCY BREAKOUT

set AH to 23H  ;CONTROL BREAKOUT FUNCTION
set AL to 0    ;ENABLE OPTION
INT 16H        ;EXECUTE FUNCTION
```

This sets the keyboard as follows:

- IBM-compatible keyboard layout
- All shift keys operational
- All shifts inactive
- [Ctrl][Alt][DEL]enabled
- Emergency breakout enabled

35.3.4 Simulating keyboard input

To insert keystrokes in the input buffer, use INT 16H function 5. This function uses the keys' scan codes, which are described in Appendix E.

This can be used for a number of purposes. For example:

- To provide default input for the current application.
- To provide deferred input, for the next application run or the MS-DOS command line.

For example, suppose an application (written in a language that supports line input) is asking for the user to input a filename. Rather than leave the user to make a up a filename it could offer a default by using the following code. The user can then accept the default by pressing [RETURN] or edit it first by using [Left] and retyping, then press [RETURN].

```
;*************************************************
;* EXAMPLE: INSERTING KEYSTROKES FROM A PROGRAM *
;*************************************************

set AM to 5              ;INSERT CHARACTER FUNCTION
set n to 0               ; CHARACTER COUNTER

LOOP:
set CH to scan code n    ;GET CHARACTER
set CL to character n
INT 16H                  ;EXECUTE FUNCTION
increment n
if more characters, goto LOOP

display prompt           ;SYNTAX DEPENDS ON LANGUAGE
display keystrokes       ;    "         "
get string               ;    "         "

return
```

To simulate use of the special **Hunter 16** control functions and keys (such as [PWR] or the LCD contrast and backlight keys), you must use a different technique. Each major control function has its own separate function. For example, to simulate use of the backlight key ([PAW][L]), use the following routines:

```
;****************************************
;* EXAMPLE: CONTROLL ING THE BACKLIGHT *
;****************************************

;BACKLIGHT ON

set AM to 64H       ;SET BACKLIGHT FUNCTION
set AL to 10H       ;MAXIMUM LEVEL IE ON
INT 10H             ;EXECUTE FUNCTION
return

;BACKLIGHT OFF

set AH to 64H       ;SET BACKLIGHT FUNCTION
set AL to 0         ;MINIMUM LEVEL IE OFF
INT 10H             ;EXECUTE FUNCTION
return
```